

Delay- and Resource-Aware Satellite UPF Service Optimization

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Abstract—Executing 5G core network functions on satellites has become crucial to enhance satellite network management and service capabilities. The User Plane Function (UPF) is responsible for efficient data traffic forwarding and is envisioned as a key and pioneering core network function that will be deployed on satellites. However, managing and providing services with satellite UPFs face dual challenges. Limited satellite resources constrain the user scale that a satellite UPF can service, resulting in an unguaranteed service delay. Moreover, the extremely rapid mobility of satellites renders it difficult for satellite UPFs to provide seamless services. To address the above challenges, this paper presents the first-of-its-kind service optimization scheme for satellite UPFs in terms of switch control, state migration, and traffic routing. To provide guaranteed service delay, we provide a theoretical analysis based on the M/G/1 queue model, demonstrating the service delay-resource consumption trade-off. A satellite UPF switch control scheme is integrated into the service optimization process, which can decrease satellite UPF service delay while saving satellite resources by adjusting the switch control parameters. To provide seamless services, we propose a satellite UPF-oriented state-aware service migration and traffic routing (UPF service optimization) algorithm. A policy network-based reinforcement learning approach is employed to dynamically perceive the satellite network's state as well as the satellite UPF switch state. Building upon the optimization of service delay through satellite UPF switch control, the processes of state-aware state migration and traffic routing are further employed to reduce delay, ensuring seamless service effectively. Experiments reveal that the proposed algorithm outperforms other benchmark algorithms under different metrics. The service delay is reduced by an average of 23.2% compared with other algorithms.

Index Terms—Satellite UPF, service optimization, delay-resource consumption trade-off, policy network.

I. INTRODUCTION

BENEFITING from the progress of next-generation mobile communication network technologies, large-scale low Earth orbit (LEO) constellations are expected to serve as both footholds and starting points for space exploration [1], [2]. B5G

and 6G envision the space-ground integrated information networks to address issues of insufficient ground network coverage and traffic supply [3]. Thus, the deployment of mobile communication networks in LEO constellations is imminent [4]. Starting from R16, 3GPP has been actively promoting the fusion of 5G and satellite communication [5], [6]. In the R18, discussions have unfolded about satellite computing and local data exchange based on the satellite user plane function (UPF) [7]. The satellite UPF will support global user data traffic routing and forwarding, accelerating satellite edge computing, and bringing resource-intensive and latency-sensitive applications from space to the public. Therefore, the research and development of satellite UPF technology are of significant importance for building the space-ground integrated information network and have garnered widespread attention.

In our previous work [8], we have validated the feasibility of satellite UPF deployment. However, enhancing the functions of UPFs onboard is non-trivial. First, the onboard resources are significantly limited. UPFs were originally designed for terrestrial mobile communication networks and demanded substantial energy and computational support. According to statistics, the servers used to deploy UPF consume > 100 W of electricity daily, and provisioning UPF services can cause server CPU utilization rates to exceed 85% [9]. Yet, to extend the service life of the battery, the discharge depth of the satellite battery usually cannot exceed 30% [10]. The consumption of satellite energy or resources by the satellite UPF needs to be strictly controlled. How to provide guaranteed UPF performance with the limited satellite resources remains an unsolved issue. Second, the high-speed movement of LEO satellites causes service discontinuity of satellite UPFs. For example, Starlink satellites move at a speed of 7.9 km/s, completing an orbit around the Earth in just 90 minutes. Users connecting through satellite UPFs may experience connection interruptions frequently. When users connect to a new satellite, the corresponding service states and context information (stored at satellite UPFs) should also be migrated to the UPF on the new satellite, and a new data path should be rapidly established to route the data traffic. How to ensure service continuity during state migration and traffic routing while conserving space resources (including satellite computation resources and inter-satellite bandwidth) needs to be addressed.

To summarize, constrained by the limited resources and highly dynamic nature of satellites, satellite UPFs cannot fully unleash their potential. On the one hand, deploying the core network on satellites in a rational manner, as well as

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configuring the core network functionality, is a challenge. Some works have begun to try to deploy the core network or UPF in non-terrestrial environments [8], [11], [12]. However, LEO satellites experience more rapid and extensive mobility than other non-terrestrial environments, such as unmanned aerial vehicles (UAVs). Besides, they ignore the in-depth optimization of core network services. On the other hand, the processes of user state migration and data traffic routing should be optimized to guarantee the satellite UPF service performance and continuity. Existing studies on satellite service migration and traffic routing can be categorized into inter-satellite migration [13], [14], [15] and satellite-ground migration [16], [17], [18]. They usually assume that the destination node for migration/routing is predetermined, which is impractical as there are usually multiple destination nodes that can be selected.

In this paper, we seek to optimize satellite UPF services taking into account the space properties of limited resources and high dynamics. Solving this problem is challenging in three folds. First, there is a trade-off between resource utilization and performance guarantee when optimizing the satellite UPF services. To minimize the service delay of satellite UPF, it is necessary to keep UPF switched on at all times. However, continuing to switch on UPF will consume a large amount of satellite resources, causing unnecessary waste. Second, the topology of the satellite networks and the available satellite resources change dynamically over time and with user requests. Static service migration and traffic routing strategies lack real-time awareness of satellite network state information. Rigid target satellite UPF selection and path planning strategies may lead to low bandwidth resource utilization and unsatisfactory delays. Last but not least, state migration, traffic routing, and satellite resource optimization are closely coupled. Satellite UPF services need to quickly establish new routing paths based on existing user states. Yet, the data forwarding service cannot be established until the new satellite UPF has loaded the user states. Additionally, the resource states of the satellite affect the decision-making process for target selection in state migration and traffic routing.

This paper presents a service optimization scheme for satellite UPFs by designing corresponding state migration, traffic routing, and satellite UPF switch control strategy. The formulation of the service optimization problem for satellite UPFs is presented as an integer linear programming problem, which is proved to be NP-hard. To provide guaranteed service delay, we identify limited satellite resources as a constraint and enable the switch control mechanism for satellite UPF. The switch control process is modeled as the M/G/1 queue model. We perform a theoretical analysis of the delay-resource consumption trade-off in satellite UPF switch control. Switch control parameters are flexibly adjusted by using the N -limited scheme. To provide seamless services, state migration, and traffic routing are coordinated to cater to satellite dynamics and maintain service continuity. We design a reinforcement learning agent based on the policy network, where the agent can extract relevant satellite network states, including information about the state of the satellite UPF switch. Furthermore, we propose a state migration and traffic routing algorithm based on satellite UPF state awareness. The

agent selects the appropriate target satellite node to perform migration and routing decisions, effectively ensuring seamless services of the satellite UPF.

The contributions of this paper are summarized as follows.

- To our best knowledge, this paper is the first to optimize the performance of satellite UPF services concerning state migration, traffic routing, and satellite UPF switch control, ensuring the continuity of satellite UPF services effectively.
- The satellite UPF service optimization problem is modeled as a 0/1 linear programming problem. On the premise of configuring the satellite UPF switch control parameters, we propose a network state-aware satellite UPF service optimization algorithm based on the policy network. The reinforcement learning agent makes state migration and traffic routing decisions by perceiving the latest network state.
- Simulations show the ideal performance of the proposed algorithm. Compared with baselines, the proposed algorithm optimizes 13.6%, 22.9%, and 14.7% on average in terms of total resource consumption, total delay, and user request acceptance rate, respectively.

The remainder of this paper is organized as follows. Section II reviews relevant studies on satellite core networks, service migration, and service switch control. In Section III, the system model is introduced. The relevant analysis of satellite UPF service optimization is provided in Section IV. Section V designs the satellite UPF service optimization algorithm based on the policy network. Section VI provides simulation experiments and result analysis. Finally, Section VII concludes this paper.

II. RELATED WORK

This paper considers user state migration, data traffic routing, and satellite UPF switch control as crucial means to optimize satellite UPF performance. Therefore, this section investigates relevant work concerning core network deployment, service migration and routing algorithms, service switch control, and network service optimization.

A. Core Network Deployment

The fusion of 5G with satellite networks has garnered considerable attention. 3GPP initiated research on non-terrestrial network projects and engaged in developing related standards, but largely focused on wireless access networks [6]. Some scholars have made preliminary attempts to verify the feasibility of deploying core networks on satellites. Xing et al. [8] conducted the first deployment of a lightweight 5G core network on a satellite. They deployed the tailored access and mobility management function, the session management function, and the UPF onto an LEO satellite successfully. Li et al. [11] discovered that deploying the 5G core network on satellites could encounter signaling storms. Consequently, they decoupled the core network functions from the state. User equipment acted as repositories to store states, and a corresponding addressing scheme was devised. Li et al. [4] proposed a cloud-edge distributed core network architecture for 6G, in which the cloud

core network coordinates various edge core networks, and the edge core networks are further decentralized to the network edges. Besides, Moradi et al. [12] deployed the core network on UAVs. It clarified the challenge of limited resources on UAV platforms and designed state migration schemes among different UAVs. However, they ignore the trade-off between resource guarantee and delay optimization.

Building upon the success of deploying core networks on satellites as mentioned above, this paper further discusses relevant issues related to the satellite UPF. Different from the above work, this paper investigates the challenges of satellite UPF service state migration and traffic routing caused by satellite motion, as well as explores how to further optimize satellite UPF services under resource-constrained conditions.

B. Service Migration and Routing

Service migration or traffic routing in terrestrial networks has been well studied. Ray et al. [19] proposed an active mechanism based on reinforcement learning to jointly solve the problem of microservice placement and service migration caused by user movement. Li et al. [20] suggested using a cloud-edge collaborative service migration strategy to enhance real-time rendering applications. By extracting multi-user layer shared information, Tang et al. [21] designed and implemented a reinforcement learning algorithm based on policy gradient to optimize the container migration delay in the edge-assisted vehicular networks. Other works [22], [23], [24] also focused on cost optimization of service migration and traffic routing in terrestrial networks.

Researchers have conducted extensive studies on satellite service migration and routing algorithms or mechanisms. Geng et al. [25] characterized the satellite network based on delay queue evaluation, deriving relevant metrics affecting delay variations, and ultimately selecting the minimum path set. However, this work did not further elaborate on how to select the final routing path from the minimum path set. Deng et al. [14] proposed an online bandwidth-aware migration algorithm to determine the optimal migration strategy, aiming to reduce latency while handling changes in satellite network topology. For the satellite-ground service migration, Cao et al. [16] designed a migration framework for satellite-ground integrated networks to jointly optimize communication and computational resources. Li et al. [18] suggested integrating satellite networks as a part of radio access networks to enhance their performance. Similarly, to enhance network capacity, Sun et al. [17] proposed a two-stage algorithm to address the service deployment and service migration issues in satellite-ground IoT. However, the algorithm assumed predictable task parameters, which did not align with the random demands of large-scale satellite-ground IoT networks.

Existing works usually design routing or migration schemes with a predetermined target node, while the target nodes are often uncertain in real conditions. We include the selection of the target satellite in the optimization process. Additionally, we do not just address traffic routing or service migration independently. Instead, we treat them as an integrated process within the context of satellite UPF.

C. Service Switch Control

In special scenarios such as satellite networks, novel resource/energy-efficiency solutions are even more needed. Due to the limited deployment of UPF on satellites, there is a lack of representative work on the UPF switch. However, efforts towards energy efficiency in network infrastructure have been ongoing for quite some time, particularly with in-depth research on base station sleep modes. Niu et al. [26], [27] initiated research on base station sleep strategies based on queueing models. Building upon their progress, Zhang et al. [28], [29] followed suit and further achieved vertical offloading of base stations in the hyper-cellular network architecture. They derived the maximum sleep probability for energy-efficient base station operations. Wu et al. [30] proposed a dynamic sleep control framework for base stations based on traffic awareness. In addition, Noh et al. [31] investigated the user-number threshold-based base station switch strategy for homogeneous/heterogeneous networks and derived the optimal user threshold for coverage guarantee. This paper seeks to apply switch control mechanisms to the unique scenario of satellite UPF, where the satellite UPF operates under strict resource constraints.

D. Learning-Based Service Optimization

Machine learning holds promise as a promising paradigm for solving network service optimization problems. Sthapit et al. [13] demonstrated the applicability of machine learning algorithms in satellite service migration and utilized them to address security risks associated with service migration. However, they do not take into consideration the impact of satellite network topology changes on service migration. Tsai et al. [32] determined the next hop for multiple requests as corresponding actions, resulting in shorter delays compared to the shortest path method. Besides, Chen et al. [33] proposed a dual-time-scale service caching, computation offloading, and resource allocation joint optimization scheme based on dual-layer deep reinforcement learning to minimize energy consumption in ultra-dense networks. However, it does not optimize the delay of ultra-dense networks to improve the quality of service (QoS). In terms of non-terrestrial network service optimization, Ning et al. [34] leveraged deep reinforcement learning algorithms to simultaneously reduce computational costs for both ground users and UAVs from the perspective of UAV trajectory optimization. Similarly, Karmakar et al. [35] designed a federated learning method for UAVs based on reinforcement learning. Through task offloading and interference suppression, the operating efficiency and safety of UAVs are significantly improved.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Problem Description and System Model

In this paper, assuming a specific number of UPFs have already been deployed in the LEO constellation. Satellite UPFs can be dynamically switched on or off based on the load. When the satellite UPF is switched on, user requests can receive a quick response without incurring additional delays. Conversely, although the satellite UPF that is switched off can conserve

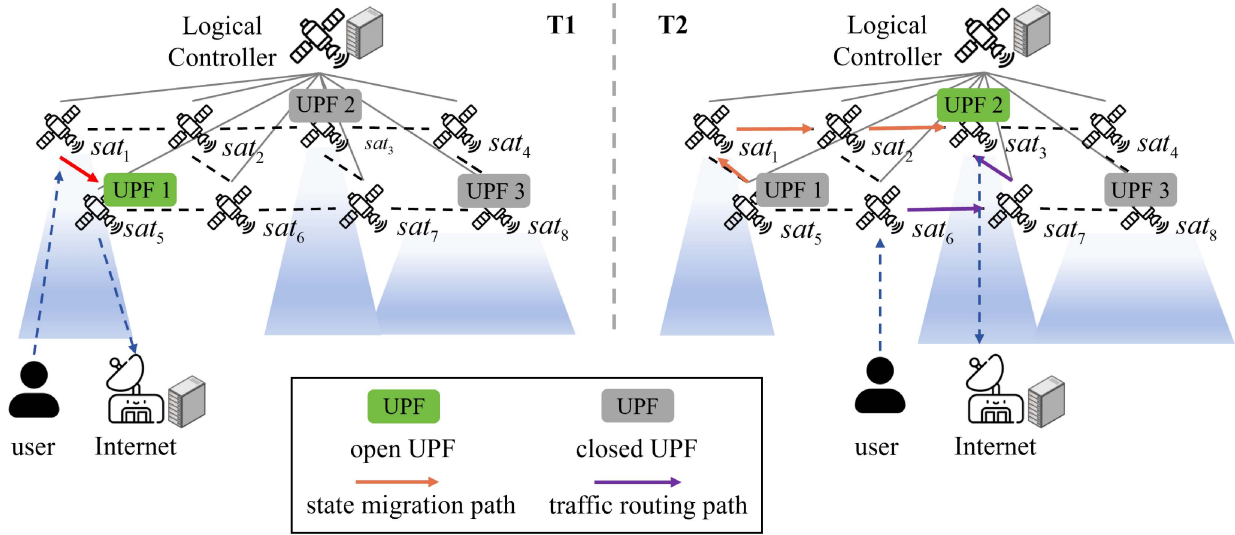


Fig. 1. Satellite UPF service optimization scenario involving state migration, traffic routing, and switch control.

satellite resources, it cannot provide timely services to incoming users. Satellite UPFs perform traffic forwarding functions and route the data packets between users and Internet/data networks based on specific policies. Moreover, satellite UPFs also support core network functions such as packet detection, execution of user plane policy rules, user plane's QoS processing, and packet labeling. When the satellite UPF serves a user request, the UPF maintains a series of packet context states, such as source address, destination address, connection identifier, QoS parameters, billing, etc.

This paper attempts to optimize the service performance of satellite UPF by solving the critical problems of state migration, traffic routing, and satellite UPF switch control. The scenario is described in Fig. 1. Multiple satellites can cover the same user at time T , and the user can access any one of the satellites. We specify that each user request consists of session-related data and state. The data is the business traffic processed by the core network, while the state is established by the user during the session. In the proposed scenario, we choose a satellite located in the center of the satellite network as a logical centralized controller, which is responsible for storing information about the satellite network and formulating service optimization strategies. In the absence of significant differences in satellite resource capacity, the central satellite is more likely to provide lower communication delay with other satellites and good redundancy [36]. On the one hand, the central satellite has good coverage and the lowest average path length to other satellites, which can significantly reduce the communication delay [37]. On the other hand, the control function can be migrated to the adjacent satellite when the original controller is unavailable, and the communication overhead between the satellite and other satellites is relatively small. The communication overhead between the controller and other satellites comes from satellite state collection and decision issuance. However, inter-satellite links usually use laser communication with a communication rate of Gbps [38], and

the state and decision information are relatively very small. The communication overhead between the controller and other satellites is acceptable. Therefore, this paper focuses on the user state migration and data traffic routing process after state collection and decision issuance.

As shown in the example in Fig. 1, a user accessed the satellite network through sat_1 at $T1$. Data traffic is routed through inter-satellite links and forwarded by satellite UPF 1 to the terrestrial Internet/data network. Due to satellite movement, the user and the Internet/data network enter the coverage of different satellites at $T2$, while the user accessed sat_6 . A new satellite UPF is needed to forward data traffic. As a result, UPF 2 is selected to provide forwarding services. Since UPF 2 was previously in a closed state, it is switched on when the user requests access. User states are migrated from UPF 1 to UPF 2 to ensure service continuity. Subsequently, a traffic routing path is established between sat_6 and sat_3 , allowing UPF 2 to continue providing data traffic forwarding services between users and the Internet/data network. After UPF 1 remains idle, it is switched off to conserve satellite resources.

We consider an LEO satellite system with a set of $S = \{1, 2, \dots, s, \dots, S\}$ satellites. Each satellite s is equipped with a specific amount of available CPU resources c_s . User requests are denoted as $R = \{1, 2, \dots, r, \dots, R\}$. In the proposed scenario, user requests are transmitted in the form of data packets within the satellite network. Simultaneously, the satellite UPF maintains user states. We use v_r^{state} and v_r^{data} to represent the size of state data packets and traffic data packets of user r . If satellites s and s' are two adjacent satellites and there is an inter-satellite link, the bandwidth of the inter-satellite link is expressed as $b_{s,s'}(t)$. Let $\alpha_{s,r}(t)$ indicate whether the user request r is accessed to satellite s at t , which is given by

$$\alpha_{s,r}(t) = \begin{cases} 1, & r \text{ is accessed to } s, \\ 0, & r \text{ is not accessed to } s. \end{cases} \quad (1)$$

B. Service Optimization Model

1) *Migration and Routing Model*: Due to the extensive movement of LEO satellites on a global scale, neither terrestrial users nor Internet/data networks can maintain a stable connection with a satellite UPF for a long period. When a satellite UPF leaves its current coverage area, it needs to migrate the relevant states to a new satellite UPF. Subsequently, new data traffic routing paths are established between the satellite accessed by users and the new satellite UPF. Therefore, we define the migration process of user states and the routing process of data traffic, respectively.

We set up a function $\beta_{s,r}(t) = 1$ to indicate the UPF deployed on satellite s to become a new access point and exert UPF-related capabilities. If $\beta_{s,r}(t) = 0$, the satellite UPF does not perform its related functions. For ease of subsequent calculations, we define a set of migration paths as follows

$$\mathcal{L}^{\text{state}} = \{\mathcal{L}_1^{\text{state}}, \mathcal{L}_2^{\text{state}}, \dots, \mathcal{L}_R^{\text{state}}\}. \quad (2)$$

Let $h_r^{\text{state}} = |\mathcal{L}_r^{\text{state}}|$, $r \in \mathcal{R}$ be the number of inter-satellite link hops in path $\mathcal{L}_r^{\text{state}}$.

The set of data traffic routing paths is given by

$$\mathcal{L}^{\text{data}} = \{\mathcal{L}_1^{\text{data}}, \mathcal{L}_2^{\text{data}}, \dots, \mathcal{L}_R^{\text{data}}\}. \quad (3)$$

Similarly, $h_r^{\text{data}} = |\mathcal{L}_r^{\text{data}}|$, $r \in \mathcal{R}$ indicates the number of inter-satellite link hops in path $|\mathcal{L}_r^{\text{data}}|$.

In the process of state migration, the total delay is related to the transmission delay and propagation delay. In the state migration path $\mathcal{L}_r^{\text{state}}$ of user request r , the transmission delay of user state on each hop is $\frac{v_r^{\text{state}}}{b_{i,i+1}(t)}$, $i = 1, 2, \dots, h_r^{\text{state}}$, where i and $i + 1$ are adjacent satellites in $\mathcal{L}_r^{\text{state}}$. $b_{i,i+1}(t)$ indicates the inter-satellite link bandwidth between satellites i and $i + 1$. Furthermore, the transmission delay of the whole state migration path $\mathcal{L}_r^{\text{state}}$ is get from $\sum_{i=1}^{h_r^{\text{state}}} \frac{v_r^{\text{state}}}{b_{i,i+1}(t)}$. Therefore, the total transmission delay for state migration of all user requests can be calculated from

$$D^{\text{state,tran}} = \sum_{t=0}^T \sum_{s=1}^S \sum_{r=1}^R \sum_{i=1}^{h_r^{\text{state}}} \beta_{s,r}(t) \frac{v_r^{\text{state}}}{b_{i,i+1}(t)}. \quad (4)$$

The propagation delay is determined by the distance between the source and destination satellites and the propagation speed. For the user request r , the propagation delay of state migration in the migration path $\mathcal{L}_r^{\text{state}}$ is $\sum_{i=1}^{h_r^{\text{state}}} \frac{F_{i,i+1}^{\text{state}}}{c}$, where $F_{i,i+1}^{\text{state}}$, $i = 1, 2, \dots, h_r^{\text{state}}$ is the distance of each hop of the state migration path. c represents the speed of light. We can get the total propagation delay for all user requests from

$$D^{\text{state,pro}} = \sum_{t=0}^T \sum_{s=1}^S \sum_{r=1}^R \sum_{i=1}^{h_r^{\text{state}}} \beta_{s,r}(t) \frac{F_{i,i+1}^{\text{state}}}{c}. \quad (5)$$

Similarly, the total delay is composed of transmission delay and propagation delay during the data traffic process. Specifically, we define the transmission delay for traffic routing of user request r in the path $\mathcal{L}_r^{\text{data}}$ as $\sum_{i=1}^{h_r^{\text{data}}} \frac{v_r^{\text{data}}}{b_{i,i+1}(t)}$, $i = 1, 2, \dots, h_r^{\text{data}}$, where i and $i + 1$ are adjacent satellites in $\mathcal{L}_r^{\text{data}}$. Based on this, the total delay for traffic routing of all user requests can be

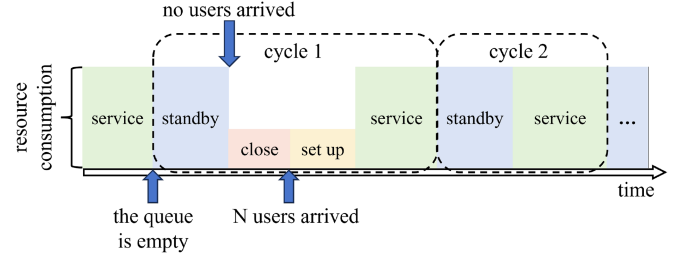


Fig. 2. UPF switch control queue based on N -limited scheme.

obtained from

$$D^{\text{data,tran}} = \sum_{t=0}^T \sum_{r=1}^R \sum_{i=1}^{h_r^{\text{data}}} \frac{v_r^{\text{data}}}{b_{i,i+1}(t)}. \quad (6)$$

The propagation delay incurred by the traffic routing of the user request r is given by $\sum_{i=1}^{h_r^{\text{data}}} \frac{F_{i,i+1}^{\text{data}}}{c}$, where $F_{i,i+1}^{\text{data}}$ represents the distance of each hop of the traffic routing path $\mathcal{L}_r^{\text{data}}$. Therefore, the total propagation delay of traffic routing can be got from

$$D^{\text{data,pro}} = \sum_{t=0}^T \sum_{r=1}^R \sum_{i=1}^{h_r^{\text{data}}} \frac{F_{i,i+1}^{\text{data}}}{c}. \quad (7)$$

In addition, the energy consumption during state migration and traffic routing are defined. We focus on the energy consumption caused by satellite UPF communications. The energy consumption of the state transition and traffic routing processes is influenced by the satellite's transmission power, which is given by

$$W = \underbrace{\sum_{t=0}^T \sum_{r=1}^R \sum_{s,s' \in \mathcal{L}_r^{\text{state}}} p_s \frac{v_r^{\text{state}}}{b_{s,s'}(t)}}_{\text{energy consumption of transmitting state}} + \underbrace{\sum_{t=0}^T \sum_{r=1}^R \sum_{s,s' \in \mathcal{L}_r^{\text{data}}} p_s \frac{v_r^{\text{data}}}{b_{s,s'}(t)}}_{\text{energy consumption of transmitting data}}. \quad (8)$$

where p_s is the transmit power of the satellite s .

2) *Switch Model*: In satellite networks, each user request arises independently and randomly with a fixed probability, similar to those in terrestrial mobile communication systems [39]. The aggregated user request arrival process can be modeled as a Poisson process when there are sufficient user requests. Moreover, we do not impose constraints on the satellite serving process. Therefore, we model the switch control process of the satellite UPF as an M/G/1 queue model, as shown in Fig. 2. The CPU resources consumed by satellite UPF during the service stage of providing services is denoted by c_u^{ser} . When the service queue of the UPF is empty, it transitions to the standby stage, which requires CPU resources of c_u^{sta} . Furthermore, a standby duration, represented as t_u^{sta} , is configured. If any user requests arrive at the satellite UPF within t_u^{sta} , the satellite UPF returns to the service stage, immediately. On the contrary, it enters a lower CPU consumption close stage, denoted as c_u^{clo} . For the N -limited scheme, the satellite UPF transitions from the close

stage to the setup stage when no less than N user requests arrive. In the setup stage, the CPU consumption is c_u^{set} . We define the duration of the setup stage as t_u^{set} .

If user requests arrive when the satellite UPF is switched off, i.e., during the close stage or the setup stage, it incurs additional delay. Reducing the duration of the close or setup stages may optimize service delay but could result in higher resource consumption. Therefore, there is a trade-off between delay and resource consumption in the satellite UPF switch control model. Adjusting the satellite UPF switch control parameters ensures low resource consumption while providing low-delay service to users. The delay and resource consumption in the process of state migration and traffic routing based on the satellite UPF can also be optimized.

The set of satellite UPFs is represented as $\mathcal{U} = \{1, 2, \dots, u, \dots, U\}$. We define a function $\gamma_u(t-1, t)$ to indicate whether the satellite UPF is switched on after the setup stage, which is given by

$$\gamma_u(t-1, t) = \begin{cases} 1, & 0 \leq \sum_r y_s \alpha_{s,r}(t-1) < N \& \sum_r l_r \alpha_{s,r}(t) \geq N, \\ 0, & \text{others,} \end{cases} \quad (9)$$

where $\&\&$ is the “AND” operation. $y_s = \{0, 1\}$ indicates whether UPF u is deployed on satellite s .

In the M/G/1 queue model, t_u^{set} follows a general distribution with a mean of $A(t_u^{\text{set}})$ and a squared coefficient of variation of $S(t_u^{\text{set}})^2$, i.e.,

$$A(t_u^{\text{set}}) = \mathbb{E}[t_u^{\text{set}}], \quad S(t_u^{\text{set}})^2 = \frac{\mathbb{E}[(t_u^{\text{set}})^2] - A(t_u^{\text{set}})^2}{A(t_u^{\text{set}})^2}. \quad (10)$$

To prove that the delay of the satellite UPF switch control process is bounded and further analyze the form of delay variation, we define the Laplace-Stieltjes transform of the standby stage duration and setup stage duration as $\tilde{t}_u^{\text{sta}}(j)$ and $\tilde{t}_u^{\text{set}}(j)$, respectively, which are calculated from

$$\tilde{t}_u^{\text{sta}}(j) = \mathbb{E}[e^{-s \cdot t_u^{\text{sta}}}], \quad \tilde{t}_u^{\text{set}}(j) = \mathbb{E}[e^{-s \cdot t_u^{\text{set}}}]. \quad (11)$$

User requests arrive following a Poisson process with a rate of λ . According to [40], [41], the additional delay caused by the introduction of non-service stages for the satellite UPF u is calculated from

$$D_u^{\text{swi}} = \frac{p^{\text{sta}}[N(N-1) + 2N\rho^{\text{set}} + (1 + S(t_u^{\text{set}})^2)\rho^{\text{set}^2}]}{2\lambda^2[p^{\text{sta}}(N + \rho^{\text{set}}) + 1 - p^{\text{sta}}]}, \quad (12)$$

where $p^{\text{sta}} = \mathbb{E}[e^{-\lambda t_u^{\text{sta}}}]$ is the probability of transitioning from the standby stage to the close stage, i.e., the close probability. $\rho^{\text{set}} = \lambda A(t_u^{\text{set}})$ denotes the request density.

In the switching cycle, each stage generates distinct resource consumption. We obtain the resource consumption of the satellite UPF u as follows

$$C_u^{\text{swi}} = \frac{1}{\lambda} \left\{ \rho c_u^{\text{ser}} + \{\mathbb{E}[L]\}^{-1} \left[\frac{c_u^{\text{clo}}(1 - p^{\text{sta}}) + N p^{\text{sta}} c_u^{\text{clo}}}{\lambda} + p^{\text{sta}} A(t_u^{\text{set}}) c_u^{\text{set}} \right] \right\}, \quad (13)$$

where $\mathbb{E}[L]$ represents the overall average duration of the switching cycle, which will be provided in Section IV-A. As the process of user requests arriving at the satellite UPF follows a Poisson distribution with λ , we have $\rho = \lambda A(t_u^{\text{ser}})$, where $A(t_u^{\text{ser}})$ denotes the mean of the service time t_u^{ser} .

Throughout the entire service request process, the additional delay caused by the satellite UPF switch is given by

$$D^{\text{swi}} = \sum_{t=1}^T \sum_{s=1}^S \gamma_u(t-1, t) D_u^{\text{swi}}. \quad (14)$$

Similarly, the additional resource consumption caused by the satellite UPF switch is

$$C^{\text{swi}} = \sum_{t=1}^T \sum_{s=1}^S \gamma_u(t-1, t) C_u^{\text{swi}}. \quad (15)$$

C. Problem Formulation

This paper focuses on the issue of service optimization of satellite UPF. The objective is to optimize the total delay and resource consumption in the process of state migration, traffic routing, and satellite UPF switch control.

The total delay of service migration and traffic routing problems of satellite UPF with switch control can be obtained by

$$\mathbf{D} = \underbrace{D^{\text{state,tran}} + D^{\text{state,pro}}}_{\text{state migration delay}} + \underbrace{D^{\text{data,tran}} + D^{\text{data,pro}}}_{\text{traffic routing delay}} + D^{\text{swi}}. \quad (16)$$

The optimization problem is defined as

$$\min \mathbf{D}$$

$$\text{s.t. } C1 : C^{\text{swi}} \leq \phi_1,$$

$$C2 : W \leq \phi_2,$$

$$C3 : \sum_{s=1}^S \beta_{s,r}(t) = 1, \quad \forall r \in \mathcal{R}$$

$$C4 : \sum_{s=1}^S \alpha_{s,r}(t) \geq 1, \quad \forall r \in \mathcal{R}$$

$$C5 : y_s \left[c_u^{\text{ser}} + \sum_{r=1}^R \beta_{s,r}(t) (c_r^{\text{state}} + c_r^{\text{data}}) \right] \leq c_s, \quad \forall u \in \mathcal{U}. \quad (17)$$

$C1$ and $C2$ ensure that the resource consumption of the satellite UPF switch and the energy consumption of the entire state migration and traffic routing process will not exceed the threshold. $C3$ specifies that the user can only be served by only one satellite UPF at t . $C4$ expresses that the user request r may access multiple hop satellites in a state migration or traffic routing path. Constraint $C5$ ensures that when the satellite UPF provides services, the CPU resources occupied by the satellite UPF and user requests do not exceed the CPU resource capacity of the satellite. The CPU resources occupied by user states and data traffic processing are represented as c_r^{state} and c_r^{data} ,

respectively. The optimization problem (17) is a 0/1 integer linear programming problem, and it is NP-hard. Because (17) can be polynomially reduced to a set cover problem, which is an NP-complete problem.

IV. PROBLEM ANALYSIS

In this section, we initially explore the trade-off between delay and resource consumption in the satellite UPF switch control process, laying the groundwork for subsequent adjustments to switch control parameters. Subsequently, satellite network-oriented satellite UPF service optimization is characterized as a reinforcement learning system.

A. Delay-Resource Consumption Trade-Off in Satellite UPF Switch Control

Exploring the mathematical relationship between delay and resource consumption aids in designing strategies for satellite UPF switch control. We define the squared coefficient of variation $S(t_u^{\text{ser}})^2$ and the Laplace-Stieltjes transform $\tilde{t}_u^{\text{ser}}(j)$ of the service duration t_u^{ser} . Next, we will demonstrate the existence of additional delay introduced by the satellite UPF switch through theoretical derivation.

The previous work [40], [41] provides the average length of the queue cycle, which is the time duration between two consecutive moments when the queue becomes empty, i.e., the average length between two standby stages. We have

$$\begin{aligned} \mathbb{E}[L] &= (1 - p^{\text{sta}}) \frac{1 - \rho + \lambda A(t_u^{\text{ser}})}{\lambda(1 - \rho)} \\ &\quad + p^{\text{sta}} \frac{N(1 - \rho) + \lambda(A(t_u^{\text{set}}) + NA(t_u^{\text{ser}}))}{\lambda(1 - \rho)} \\ &= \frac{1 - p^{\text{sta}} + p^{\text{sta}}(N + \rho^{\text{set}})}{\lambda(1 - \rho)}. \end{aligned} \quad (18)$$

Please refer to Appendix A, available online, for the detailed derivation of $\mathbb{E}[L]$.

Furthermore, the Laplace-Stieltjes transform of the overall sojourn time of user requests at satellite UPF is obtained from

$$\begin{aligned} t_u^{\text{tot}}(j) &= \frac{\tilde{t}_u^{\text{ser}}(j)}{\mathbb{E}[L]} \left\{ \frac{1 - p^{\text{set}}}{\lambda} + \frac{p^{\text{sta}}[1 - \tilde{t}_u^{\text{set}}(j)(\tilde{t}_u^{\text{ser}}(j))^N]}{j - \lambda + \lambda \tilde{t}_u^{\text{ser}}(j)} \right. \\ &\quad + p^{\text{sta}} \frac{\tilde{t}_u^{\text{sta}}(s) \{ [\lambda/(j + \lambda)]^N - [\tilde{t}_u^{\text{ser}}(j)]^N \}}{\lambda^2/(j + \lambda) - \lambda \tilde{t}_u^{\text{ser}}(j)} \\ &\quad \left. \frac{(1 - p^{\text{sta}})(1 - \tilde{t}_u^{\text{ser}}(j))}{j - \lambda(1 + \tilde{t}_u^{\text{ser}}(j))} \right\}. \end{aligned} \quad (19)$$

Taking the derivative of (19) with respect to j and substituting (18) into the derivative result with $j = 0$, we have

$$\begin{aligned} D_u^{\text{swi}} &= A(t_u^{\text{ser}}) + \frac{\rho(1 + S(t_u^{\text{ser}})^2)A(t_u^{\text{ser}})}{2(1 - \rho)} \\ &\quad + p^{\text{sta}} \frac{N(N - 1 + 2\rho^{\text{set}}) + (1 + S(t_u^{\text{ser}})^2)\rho^{\text{set}^2}}{2\lambda[p^{\text{sta}}(N + \rho^{\text{set}}) + 1 - p^{\text{sta}}]}, \end{aligned} \quad (20)$$

where the first two items are the average service time of satellite UPF for the normal M/G/1 queue without the standby stage, close stage, and setup stage. An additional delay is introduced by the standby, close, and setup stages, as shown in the third term. Therefore, $t_u^{\text{ser}} = 0$, $t_u^{\text{sta}} = 0$. Since (20) does not involve t_u^{sta} , by setting $t_u^{\text{ser}} = 0$, we have

$$D_u^{\text{swi}} = p^{\text{sta}} \frac{N(N - 1 + 2\rho^{\text{set}}) + (1 + S(t_u^{\text{ser}})^2)\rho^{\text{set}^2}}{2\lambda[p^{\text{sta}}(N + \rho^{\text{set}}) + 1 - p^{\text{sta}}]}, \quad (21)$$

which is identical to (12). Therefore, there is an additional delay caused by the switch of the satellite UPF.

Then, we will explore the trade-off between delay and resource consumption in the switch control process of the satellite UPF. Based on (20), we derive the average sojourn time of user requests in the satellite UPF as follows

$$\begin{aligned} D_u^{\text{swi}} &= \frac{2t_u^{\text{ser}}(1 - \rho) + (1 + S(t_u^{\text{ser}})^2)\rho t_u^{\text{ser}}}{2(1 - \rho)} \\ &\quad + \frac{N(N - 1) + 2N\rho^{\text{set}} + \rho^{\text{set}^2}(1 + S(t_u^{\text{set}})^2)}{2\lambda[N + \rho^{\text{set}} + 1/p^{\text{sta}} - 1]}. \end{aligned} \quad (22)$$

As the delay in the satellite UPF switching cycle is caused by the close and setup stages, we simplify (22) to obtain

$$D_u^{\text{swi}} = \frac{N(N - 1) + 2N\rho^{\text{set}} + \rho^{\text{set}^2}(1 + S(t_u^{\text{set}})^2)}{2\lambda[N + \rho^{\text{set}} + 1/p^{\text{sta}} - 1]}, \quad (23)$$

from which it can be observed that D_u^{swi} is monotonically increasing with respect to p^{sta} .

Similarly, by averaging the resource consumption of the satellite UPF in service stage (c_u^{ser}), standby stage (c_u^{sta}), close stage (c_u^{clo}), and setup stage (c_u^{set}), we can get (13). We denote the modified expression of (13) as

$$\begin{aligned} C_u^{\text{swi}} &= (1 - \rho)c_u^{\text{clo}} + \rho c_u^{\text{ser}} \\ &\quad + \frac{(1 - \rho)\{(1 - p^{\text{sta}})[c_u^{\text{sta}} - c_u^{\text{clo}}] + p^{\text{sta}}\rho^{\text{set}}[c_u^{\text{set}} - c_u^{\text{clo}}]\}}{1 + p^{\text{sta}}(N + \rho^{\text{set}} - 1)}. \end{aligned} \quad (24)$$

from which it can be observed that C_u^{swi} is monotonically decreasing with respect to p^{sta} .

By combining (23) and (24) and substituting C_u^{swi} , we can eventually express C_u^{swi} as follows

$$\begin{aligned} C_u^{\text{swi}} &= \rho c_u^{\text{ser}} + (1 - \rho)c_u^{\text{sta}} \\ &\quad + \frac{(1 - \rho)(2\lambda D_u^{\text{swi}} - 2\rho) + \rho^2(1 + S(t_u^{\text{ser}})^2)}{N(N - 1) + \rho^{\text{set}^2}(1 + S(t_u^{\text{set}})^2) + 2N\rho^{\text{set}}} \\ &\quad \times [\rho^{\text{set}}(c_u^{\text{set}} - c_u^{\text{clo}}) - (N + \rho^{\text{set}})(c_u^{\text{sta}} - c_u^{\text{clo}})], \end{aligned} \quad (25)$$

from which can be observed that the resource consumption C_u^{swi} and average delay D_u^{swi} in the satellite UPF switch control process have a linear relationship with the given parameters.

B. Policy Network Based Reinforcement Learning System for Satellite UPF Service Optimization

We propose a reinforcement learning-based service optimization scheme for onboard UPF that enables state migration and traffic routing of each user request. The state, action, and reward in reinforcement learning are described as follows.

State: The state $\mathbf{S}(t)$ is a description of the satellite network environment where satellite UPFs are deployed. At t , the satellite network's state is represented as $\mathbf{S}(t) = \{\nu_1(t), \nu_2(t), \dots, \nu_U(t), \dots, \nu_U(t)\}$, where $\nu_u(t)$ signifies a feature vector indicating the state of the satellite UPF u . Concretely, the feature vector for u is given by $\nu_u(t) = \{c_u(t), b_u^{\text{sum}}(t), f_u^{\text{avg}}(t), g_u^{\text{swi}}(t)\}$, with each element being normalized. $c_u(t)$ represents the available CPU resources of satellite deployed with UPF u . Besides, $b_u^{\text{sum}}(t)$ indicates the total bandwidth of the interconnecting inter-satellite links associated with the satellite where the UPF u is deployed. The average distance to other satellite UPFs is $f_u^{\text{avg}}(t)$, which can be calculated from

$$f_u^{\text{avg}}(t) = \frac{\sum_{u' \in \mathcal{U}, u' \neq u} \mathcal{D}(u, u')}{U}, \quad (26)$$

where $\mathcal{D}(u, u')$ is the shortest distance between onboard satellites u and u' , which is get from the Floyd algorithm. It means that selecting a satellite with satellite UPF with a shorter average distance as the target node is essential for ensuring low latency. Additionally, the satellite UPF's switch state $g_u^{\text{swi}}(t)$ is incorporated.

Action: In the context of service optimization of satellite UPF, the objective is to choose a suitable target node and create the delay-shortest migration/routing path. Therefore, the action is denoted as $\mathbf{A}(t) = \{u_1(t), u_2(t), \dots, u_r(t), \dots\}$.

Reward: Large or positive rewards encourage the agent to continue exploring effective actions, while small or negative rewards indicate that the agent should adjust its actions. In the described scenario, the reward is influenced by factors such as delay, resource consumption, and the switch state of satellite UPF. Therefore, the reward is expressed as

$$\mathbf{R}(t) = \begin{cases} \eta_1 \frac{1}{\sum_t \sum_r \mathcal{D}_r(t)} + \eta_2 \frac{1}{\sum_t \sum_r W_r(t)}, \\ \forall u \in \mathcal{U} \quad \gamma_u(t-1, t) = 0, \\ \eta_1 \frac{1}{\sum_t \sum_r \mathcal{D}_r(t)} + \eta_2 \frac{1}{\sum_t \sum_r W_r(t)} \\ - \eta_3 \sum_t \sum_u C_u^{\text{swi}}(t), \\ \exists u \in \mathcal{U} \quad \gamma_u(t-1, t) = 1, \end{cases} \quad (27)$$

where η_1 - η_3 are weight factors and all values are normalized. $\mathcal{D}_r(t)$ and $W_r(t)$ represent the total delay and total energy consumption for serving user request r , respectively. It reflects that choosing a satellite UPF with different switch states will have a significant impact on rewards.

Reinforcement learning is well-suited for solving decision problems in high-dimensional spaces [42], [43]. We use a policy network as the intelligent agent in this system. The satellite network provides relevant states of satellite UPFs to the intelligent agent. When the state migration and traffic routing event is

triggered, a new satellite UPF needs to be selected for data forwarding. However, the resource capacity of satellites is limited and can only serve a finite number of user requests. Additionally, different migration and routing paths can result in differentiated delay interference. Especially when the satellite UPF is switched off, it will introduce additional delays. Therefore, it is essential to gather information about the state of the satellite UPF and the satellite network in advance. As mentioned before, we have fine-grainedly extracted 4 general attributes to represent the state information of the satellite and further formalize them into a vector $\nu_u(t) = [c_u(t), b_u^{\text{sum}}(t), f_u^{\text{avg}}(t), g_u^{\text{swi}}(t)]$, $u \in \mathcal{U}$ for each satellite UPF.

We aggregate the feature vectors of all satellite UPFs into a state matrix $\mathcal{M}$ to reflect the satellite network state at t , which can be expressed as

$$\mathcal{M}(t) = [\nu_1(t), \nu_2(t), \dots, \nu_U(t), \dots, \nu_U(t)] \\ = \begin{bmatrix} c_1(t) & b_1^{\text{sum}}(t) & f_1^{\text{avg}}(t) & g_1^{\text{swi}}(t) \\ c_2(t) & b_2^{\text{sum}}(t) & f_2^{\text{avg}}(t) & g_2^{\text{swi}}(t) \\ \dots & \dots & \dots & \dots \\ c_u(t) & b_u^{\text{sum}}(t) & f_u^{\text{avg}}(t) & g_u^{\text{swi}}(t) \\ \dots & \dots & \dots & \dots \\ c_U(t) & b_U^{\text{sum}}(t) & f_U^{\text{avg}}(t) & g_U^{\text{swi}}(t) \end{bmatrix}. \quad (28)$$

When the state migration and traffic routing occur, the reinforcement learning agent perceives the network state from the satellite network. To make intelligent decisions, we employ a self-built policy network as the agent, which is depicted in Fig. 3.

The policy network consists of four layers. In addition to the input layer and the output layer, the middle two layers are the calculation layer and the softmax layer. The calculation layer applies convolution operations to each vector $\nu_u(t)$, extracting state information from the satellite network and obtaining the vector of available resources θ_u . The softmax layer converts the convolution results into probabilities, representing the probability that a satellite deployed with satellite UPF is selected as the target migration node. As the intelligent agent, the policy network takes the state matrix $\mathcal{M}(t)$ of the LEO satellite network as input. It generates a set of satellite nodes deploying satellite UPF, with the probability of being selected.

V. ALGORITHM DESIGN

The service optimization problem of the satellite UPF consists of three sub-processes: service state migration, data traffic routing, and satellite UPF switch control. In the state migration and traffic routing problem, we transform it into solving the problem of target satellite UPF selection and path planning. The objective is to guarantee service continuity and minimize delay while optimizing resource consumption to ensure QoS. For satellite UPF switch control, we adjust the switch control parameters to guide the satellite UPF switch in an offline manner. In this section, we implement a state-aware satellite UPF service optimization algorithm based on the policy network.

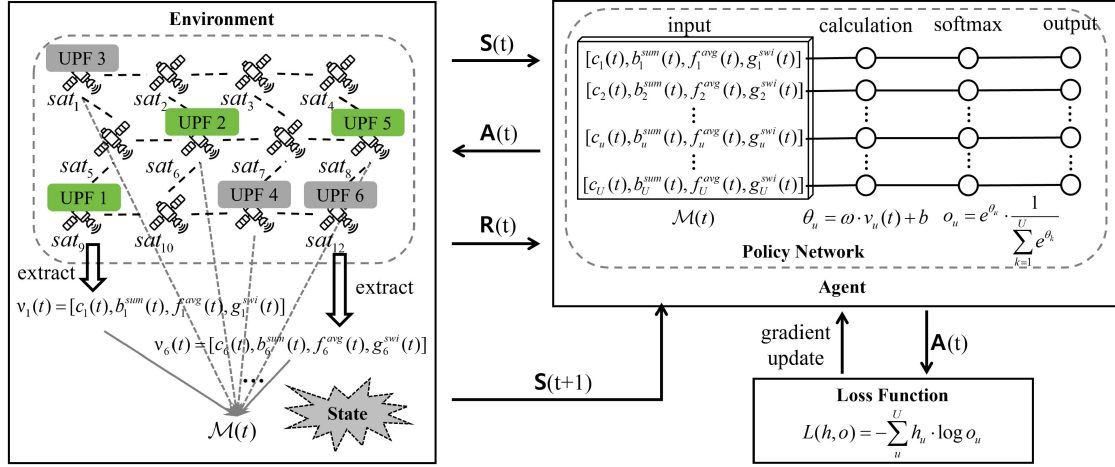


Fig. 3. Satellite UPF service optimization framework based on the policy network.

A. Dynamic Adjustment of Satellite UPF Switch Control Parameters

In Section IV-A, we have demonstrated the existence of a trade-off between delay and resource consumption in the satellite UPF switch control process. We adopt the N -limited scheme as the switch-on strategy of satellite UPF. Specifically, we consider adjusting the standby duration t_u^{sta} and parameter N to explore the optimal satellite UPF switch strategy. For t_u^{sta} : If t_u^{sta} is too short, the satellite UPF will enter the close state quickly, significantly reducing resource consumption but introducing an additional delay. If t_u^{sta} is too long, even though incoming user requests will receive immediate service, it is not conducive to resource conservation. For N : If N is too small, it means that the satellite UPF will be switched on as soon as a small number of user requests arrive, reducing delay but increasing resource consumption. Conversely, increasing N saves resources but increases delay. Therefore, we determine t_u^{sta} and N by offline calculations to strike a trade-off between delay and resource consumption in the satellite UPF switch control process.

We adjust the parameters t_u^{sta} and N based on the theoretical derivation in Section IV-A. When a smaller delay and resource consumption are obtained, t_u^{sta} and N are updated. The process of the dynamic adjustment strategy of satellite UPF switch control parameters is shown in Algorithm 1. The input includes the satellite network with UPF deployment and user requests. The parameters t_u^{sta} and N are set according to the delay-resource consumption trade-off in satellite UPF switch control. The intelligent agent need to assess the switch state of each satellite UPF. If the target satellite UPF is in the service or standby stage, the agent proceeds with the migration to that satellite UPF. If the target satellite UPF is in the close or setup stage, corresponding penalties are applied.

B. State-Aware Satellite UPF Service Optimization Algorithm

The agent autonomously perceives the state of the satellite network and achieves satellite UPF service optimization by state

Algorithm 1: Dynamic Adjustment Algorithm of Satellite UPF Switch Control Parameters.

Input: $u \in \mathcal{U}$, switch control parameters, iterations I ;
Output: t_u^{sta}, N ;

- 1: **while** $u \in \mathcal{U}$ **do**
- 2: randomly initialize t_u^{sta}, N ;
- 3: **if** u is in service or standby stage **then**
- 4: continue;
- 5: **end if**
- 6: **if** u is in close or setup stage **then**
- 7: calculate D_u^{sta} by (12);
- 8: calculate C_u^{sta} by (13);
- 9: **while** $i < I$ **do**
- 10: adjust t_u^{sta}, N ;
- 11: calculate $D_u^{\text{sta}'}$ by (12);
- 12: calculate $C_u^{\text{sta}'}$ by (13);
- 13: **if** $D_u^{\text{sta}'} < D_u^{\text{sta}} \parallel C_u^{\text{sta}'} < C_u^{\text{sta}}$ **then**
- 14: adjust t_u^{sta}, N ;
- 15: **end if**
- 16: **end while**
- 17: **end if**
- 18: optimal t_u^{sta}, N ;
- 19: **end while**

migration and traffic routing. The framework of the state-aware satellite UPF service optimization algorithm based on the policy network is shown in Fig. 3. The global movement of LEO satellites triggers the state migration and traffic routing process of user request r . The agent extracts the state matrix $\mathcal{M}(t)$ from the satellite network, which is equivalent to sensing the environmental state $\mathbf{S}(t)$. The policy network generates a set of probabilities for available satellite UPF nodes along with corresponding probabilities. The agent selects a satellite UPF as the destination node for state migration and traffic routing based on probability. Subsequently, relevant state migration and traffic routing decisions are executed. This means that the agent imposes an action $\mathbf{A}(t)$ on the environment. After that,

the satellite network will feed back an instant reward $\mathbf{R}(t)$ to the agent according to the resource consumption cost and the delay cost of the operation. Subsequently, the agent adjusts its satellite UPF selection strategy based on the received reward $\mathbf{R}(t)$, aiming to maximize the cumulative reward. Following the action $\mathbf{A}(t)$, the satellite network transitions to a new state $\mathbf{S}(t+1)$.

Algorithm 2 presents the pseudo-code for implementing the proposed algorithm, which consists of two stages: offline training and online testing. We first set the switch control parameters of the satellite UPF according to Algorithm 1. During the training, as the policy network parameters are randomly initialized, it cannot directly select the satellite UPF with the highest probability as the target migration node. There is a possibility of better target nodes. Therefore, based on the probability distribution of satellite UPF output by the policy network, we perform sampling and randomly select one as the target satellite UPF until all user requests for that iteration are allocated. We train the policy network model using policy gradients. Assuming the currently selected target satellite s is the optimal decision, we calculate the cross-entropy loss for it as follows

$$L(h, o) = - \sum_u h_u \log o_u, \quad (29)$$

where h_u is an U -dimensional hand-crafted label vector. If the u th satellite is selected, the u th position in h_u is 1, and all other positions are 0. Then, we use backpropagation to update the parameter gradients of the policy network. After the target node is selected, a breadth-first search strategy is used for state migration and traffic routing. During the testing phase, we directly utilize the trained policy network parameters to select the target satellite UPFs and use the breadth-first search algorithm to derive the migration and routing paths.

Overall, after a state migration event is triggered, the source satellite for the user state migration and the source satellite for the traffic routing are identified. The algorithm performs target satellite node selection for incoming user requests by extracting the state matrix $\mathcal{M}(t)$. Then, it generates migration paths for the state migration and the traffic routing, respectively. Once the whole process for user request r is completed, the algorithm calculates the immediate reward for the service optimization strategy. Finally, the policy network adjusts its network parameters based on the observed reward effects. In the testing phase, the algorithm takes into account the dynamic changes in the satellite network topology during the arrival and execution of user service requests. For each user request, the state matrix is re-extracted to ensure that the agent develops satellite UPF selection strategies promptly and accurately. After selecting the target satellite UPF according to the maximum probability, the shortest delay paths are planned for state migration and traffic routing.

C. Computational Complexity

The computational complexity comes from the dynamic adjustment of satellite UPF switch control parameters and the

Algorithm 2: State-Aware Satellite UPF Service Optimization Algorithm.

Input: $\mathcal{S}, \mathcal{U}, \mathcal{R}, num_epoch, batch_size$;
Output: optimized parameters, evaluation indicators;
1: **off-line:**
2: set satellite UPF switch control parameters based on algorithm 1;
3: randomly initializes policy network parameters;
4: **while** $epoch \leq num_epoch$ **do**
5: **for** $r \in \mathcal{R}$ **do**
6: $i = 0$;
7: extract state matrix $\mathcal{M}_S(t)$;
8: **for** $u \in \mathcal{U}$ **do**
9: calculate available nodes and probabilities by
 $o_u = e^{v_u} \cdot \frac{1}{\sum_{k=1}^U v_k}$;
10: **end for**
11: sampling based on o_u ;
12: calculate the loss by $L(h, o) = - \sum_u h_u \cdot \log o_u$;
13: calculate the gradient;
14: BFS-based shortest delay path selection for state;
15: BFS-based shortest delay path selection for traffic;
16: calculate instant reward by (27);
17: $i++$;
18: **if** $batch_size$ **then**
19: update parameters of policy network;
20: **end if**
21: **end for**
22: $epoch++ = 1$;
23: **end while**
24: **on-line:**
25: set satellite UPF switch control parameters based on algorithm 1;
26: update parameters based on training results;
27: **while** $r \in \mathcal{R}$ **do**
28: extract state matrix $\mathcal{M}(t)$;
29: **for** $u \in \mathcal{U}$ **do**
30: calculate the target node with the maximum probability;
31: **end for**
32: BFS-based shortest delay path selection for state;
33: BFS-based shortest delay path selection for traffic;
34: **end while**

training and testing of the satellite UPF service optimization algorithm. In the parameter adjustment process, the computational complexity is $O(UI)$, where U and I are the number of satellite UPFs and iterations of control parameter update, respectively. During the training phase, the complexity arises from state matrix processing ($O(UM)$), node selection ($O(U)$), and routing path determining ($O(S+E)$), with the sum of $O(HR(UM+U+S+E))$. Here, H is the number of training epochs. R represents the number of user requests. M denotes the dimension of the state matrix. S and E are the numbers of satellite nodes and inter-satellite links. During the testing phase, the computational complexity is $O(R(UM+U+S+E))$. $O(UM)$ is the

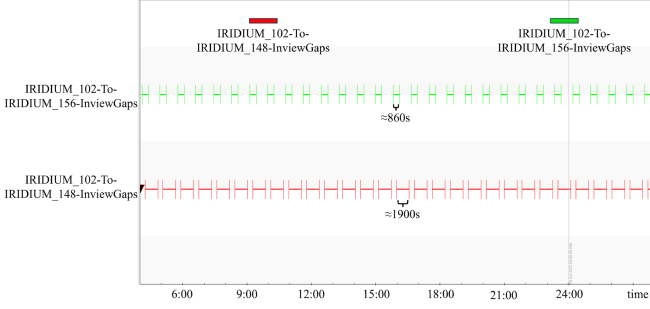


Fig. 4. The connection time between a satellite in the Iridium constellation and two satellites in adjacent orbits.

computational complexity of state matrix processing. $O(U)$ and $O(S + E)$ represent the complexity of satellite node selection and routing, respectively. To sum up, the overall complexity of the proposed solution is $O(UI + HR(UM + U + S + E) + R(UM + U + S + E))$. After simplification, the computational complexity is $O(U + R(UM + U + S + E))$.

VI. NUMERICAL RESULTS

In this section, we conduct extensive simulation experiments to evaluate the proposed algorithm. We build the policy network based on the Tensorflow. A LEO satellite network topology is created based on the Iridium constellation¹. In the performance comparison, we first compare the proposed algorithm with baselines in terms of resource consumption, service delay, etc. The impact of different network parameters on the algorithm performance is explored. We also conduct satellite UPF switch control parameter verification experiments, which further confirm that reasonable switch control parameters can optimize resource consumption while ensuring service delay.

A. Evaluation Setup

We use STK² to simulate the Iridium constellation and its dynamics, which consists of 6 near-Earth orbits with a total of 66 uniformly distributed satellites, with 6 satellites equipped with UPF deployed in each orbital plane. The Iridium constellation follows the “+Grid” connection mode, where each satellite is connected to its two adjacent satellites in the same orbit and the two closest satellites in the adjacent orbit. To simulate the dynamic nature of the constellation, we measure the duration of the connections between the satellites with two adjacent satellites in adjacent orbits of the Iridium constellation. Fig. 4 shows the access time slots between satellite IRIDIUM-102 and two neighboring satellites (IRIDIUM-148 and IRIDIUM-156). We consider the network topology of the Iridium constellation to be relatively stable within the shorter time slot ($< 860s$).

At any given time, assuming that user requests from the ground randomly access any satellite within their line of sight. We only focus on the routing process of user requests in satellite

TABLE I
PARAMETER SETTING

Parameters	Value
number of satellites S	66
number of orbit planes	6
number of UPFs U	36
CPU of satellites c_s	[100,200] TFLOPS
bandwidth of inter-satellite links $b_{s,s'}(t)$	[1,10] Gbps
number of user requests R	6000
CPU requirements of state data packets c_r^{state}	[1,10] TFLOPS
CPU requirements of traffic data packets c_r^{data}	[1,50] TFLOPS
the size of state data packets v_r^{state}	[0.05,0.5] Mb
the size of traffic data packets v_r^{data}	[0.5,5] Mb
CPU requirements of satellite UPF c_u^{ser}	30 TFLOPS
resource consumption during standby c_u^{sta}	20 TFLOPS
resource consumption during setup c_u^{set}	20 TFLOPS
resource consumption during close c_u^{clo}	5 TFLOPS
service duration $A(t_u^{ser})$	5 ms

networks, while ignoring ground-to-satellite and satellite-to-ground transmission. The data traffic is forwarded by the satellite UPF. However, due to satellite movement, it is necessary to search for new satellite UPFs to continue forwarding data traffic. The premise is that the new satellite UPF maintains the previous service states. Simulation experiments verify the forwarding performance of service states and data traffic in satellite networks. User requests come with certain resource demands. The average user request arrival rate is 40 per second. Table I lists the main simulation parameters.

B. Performance Comparison

We compare the state-aware satellite UPF service optimization algorithm based on the policy network with three other baselines that can be seamlessly applied to inter-satellite service migration scenarios.

Greedy Algorithm: Services always migrate from the current satellite UPF to the satellite UPF with the fewest inter-satellite link hops.

Random Algorithm: Services randomly select one satellite UPF from other satellite UPFs and migrate using the shortest delay path methods.

Deep Reinforcement Learning-based Online Routing Algorithm (DRL-OR) [44]: An online routing algorithm based on deep reinforcement learning for multiple agents. It deploys an agent on each routing node to make next-hop routing decisions in turn, resulting in an exponential growth of the action space. Agents need to interact with each other to obtain states such as network topology, link capacity, loss rate, etc.

Delay-sensitive Virtual Network Embedding base on Deep Reinforcement Learning (DDRL-VNE) [45]: It solves the traffic routing problem from the perspective of virtual network embedding, where network links have fixed delays. The algorithm selects the target node regardless of its load. Moreover, the shortest path algorithm based on hop is used to develop routing strategies.

1) Training Results: The data set used for simulation experiments contains 6,000 user requests, 3,000 of which are used as the training set, while the remaining are the test set. During the training phase, gradient descent is employed for training

¹<https://www.iridium.com/>

²<https://www.agi.com/>

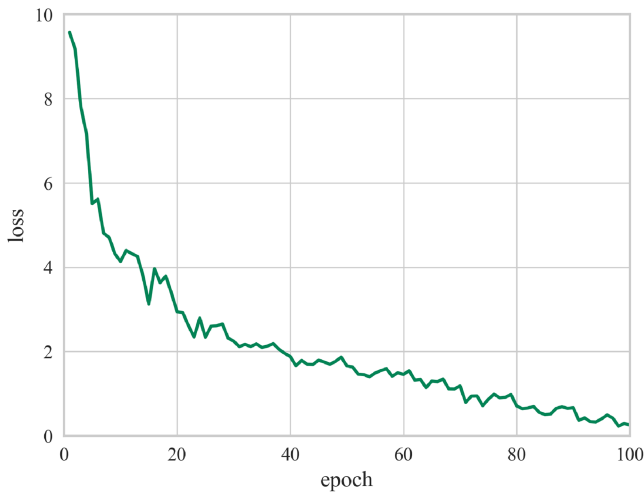


Fig. 5. Training results of loss.

the agent over 100 epochs. The policy network's parameters are updated in a batch-wise manner. We investigate the changes in cross-entropy loss during the training phase, as depicted in Fig. 5. During the training process of 100 epochs, it is obvious that the loss continues to decrease. In the later stages of training, the loss tends to stabilize at a lower level.

2) Test Results:

a) Comparison with baselines: We run the proposed algorithm on the test set and compare it with baselines. Specifically, the changes over time of the three indicators of total resource consumption, total delay, and user request acceptance rate are compared, as shown in Fig. 6.

Fig. 6(a) shows that the total resource consumption of the satellite network increases over time. As the number of user requests increases, the cumulative occupied computing resources and network resources also increase significantly. The proposed algorithm utilizes the reinforcement learning agent to perceive states of the satellite network when a user request arrives. Through flexible satellite UPF switch control and target satellite UPF selection strategy, the user states and data traffic are reasonably migrated to the target satellite UPF, making efficient use of satellite resources. The DRL-OR algorithm takes the local states of each node, including traffic features, topological information, and resource utilization, as input to infer the probability of selecting the next-hop node. Due to the limitation of deciding only the next hop node, it lacks long-term resource optimization capabilities when planning multi-hop routing paths. The DDRL-VNE algorithm preferentially selects physical nodes that are closer to the source node as the target node, ignoring the load of the target node. In addition, this method is prone to excessive resource fragmentation, leading to resource waste. The greedy algorithm selects the nearest satellite UPF as the migration node, ignoring the resource situation of the target satellite. The random algorithm also does not take the resource capabilities provided by the target satellite as a judgment condition, and the inter-satellite links are not well planned, resulting in high resource consumption. However, as time and the number of

user requests continue to grow, the performance gap between different algorithms becomes more and more obvious. The proposed algorithm achieves the lowest total resource consumption compared to other baselines. The total resource consumption of the proposed algorithm is on average 37.6% lower than other them.

Fig. 6(b) depicts the comparative results of the total delay. The proposed algorithm perceives available resources at the target satellite UPF and inter-satellite link states to minimize delays while exploring the target migration node. The DRL-OR algorithm is a distributed machine learning algorithm. Agents deployed on various nodes need frequent communication to exchange network information, resulting in additional delays and resource overhead. Additionally, the DRL-OR algorithm only makes single-hop routing decisions for user requests, lacking a global perspective. Agents need to interact frequently to obtain network states and develop local strategies, incurring additional delays. In the DDRL-VNE algorithm, the deep reinforcement learning agent extracts the delays of links adjacent to physical nodes. It only considers the optimal delay between the current node and the next-hop node, which may not be optimal for long-term optimization. The greedy and random algorithms target immediate service migration success without considering the satellite network's resource states, making them less friendly for long-term optimization focused on delay minimization. Similarly, the performance disparities among different algorithms gradually manifest as the experiments unfold. Continuously arriving user requests occupy more and more satellite resources. State migration and traffic routing decisions are made under strict resource constraints, resulting in increasing total delay. On average, the proposed algorithm reduces the delay by 23.2% compared to the other algorithms.

Fig. 6(c) shows the performance of the five algorithms on user request acceptance rate. On the whole, the performance of all the algorithms shows a trend of gradual decline. In the early stage, the satellite network resources are relatively sufficient, and the satellite UPF can provide services for most users. As user requests continue to come, the number of available satellite UPFs and satellite resources are reduced, and the number of users who cannot obtain satellite UPF services increases. The proposed algorithm utilizes the perception ability of the reinforcement learning agent to perceive the states of satellite network resources by extracting the state matrix. It can dynamically make migration decisions. The DRL-OR algorithm also uses intelligent agents to perceive local network states and obtain a slightly worse user request acceptance rate. The DDRL-VNE algorithm prioritizes delay as the optimization objective. Both hop-based shortest path and centralized selection of target nodes result in underutilization of resources, leading to a decrease in the success rate of user requests. The greedy algorithm can work well when satellite network resources are sufficient. However, when the nearby satellite UPF resources are insufficient, the performance of the algorithm will drop significantly. Similarly, the random algorithm is likely to cause too much resource fragmentation, leading to a large number of user requests that cannot be satisfied. Overall, the user request acceptance rate is increased by an average of 14% compared with the other algorithms.

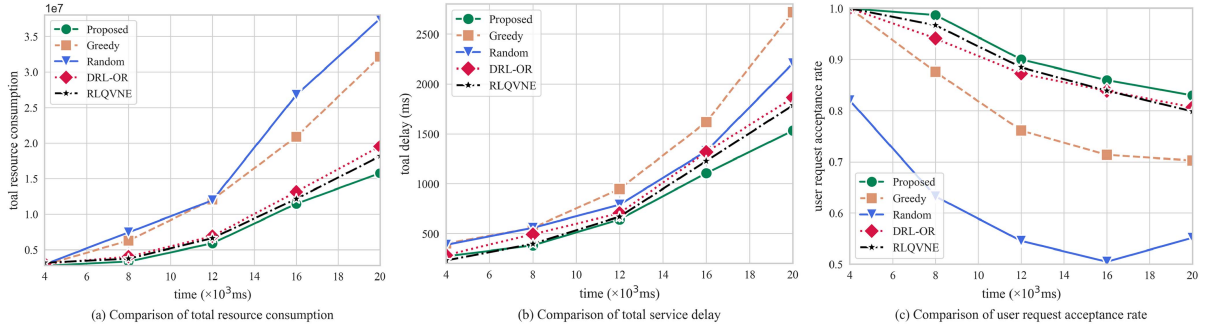


Fig. 6. Performance comparison with baselines in terms of total resource consumption, total delay, and user request acceptance rate.

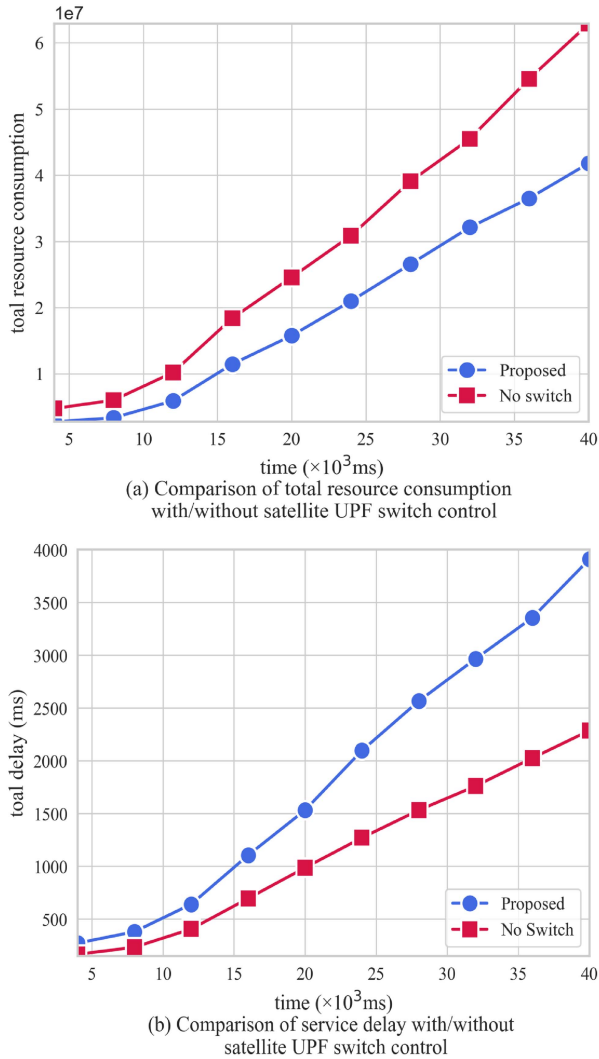


Fig. 7. Performance comparison with/without satellite UPF switch control in terms of total resource consumption and total delay.

b) Comparison with/without switch control mechanism: To verify the benefits brought by the satellite UPF switch control strategy to the proposed algorithm, we compare the performance of the algorithm with that without using the switch control scheme, and the results as shown in Fig. 7.

The experimental results effectively confirm the prior inferences. In the absence of the switch control strategy, the satellite UPF operates continuously. It ensures that users receive immediate services after accessing satellite UPF, resulting in a substantial reduction in delay. However, the persistent operation of the satellite UPF comes at the cost of relatively high resource consumption, significantly amplifying the overall resource usage. The proposed algorithm uses the N -limited strategy to guide satellite UPF switch control, effectively balancing resource consumption and service delay. Especially in extremely resource-constrained scenarios such as satellite networks, we believe that incorporating essential resource-saving mechanisms is imperative.

c) Comparison with different network conditions: In this experiment, we test the algorithm's performance under different satellite network topologies. Specifically, we observe the algorithm performance in 4-time slots. Fig. 8 illustrates the performance comparison of the three metrics in different slots.

In different time slots, satellite network topology changes significantly. Decisions regarding state migration and traffic routing require timely awareness of the new satellite network state. When state migration is triggered, the proposed algorithm dynamically extracts the state matrix from the satellite network to obtain the latest state of the satellite UPFs. Fig. 8 illustrates that the performance of different algorithms varies across different time slots for the same metric. This phenomenon arises because the connectivity between different satellites changes, leading to variations in the distance of state migration and traffic routing between satellite UPFs. Furthermore, the amount of network resources occupied by user requests also fluctuates accordingly. Overall, the performance of the proposed algorithm outperforms the other three algorithms. Therefore, the proposed algorithm maintains stability in dynamic network environments and demonstrates excellent applicability.

C. Impact of Different Switch Control Parameters

We intuitively demonstrate the effects of different switch control parameters through experimental results. Differentiated switch control parameters have a significant impact on the resource consumption and service delay of satellite UPFs. This paper employs the N -limited scheme as the switch mode for satellite UPFs. In the close stage, the transition to the next

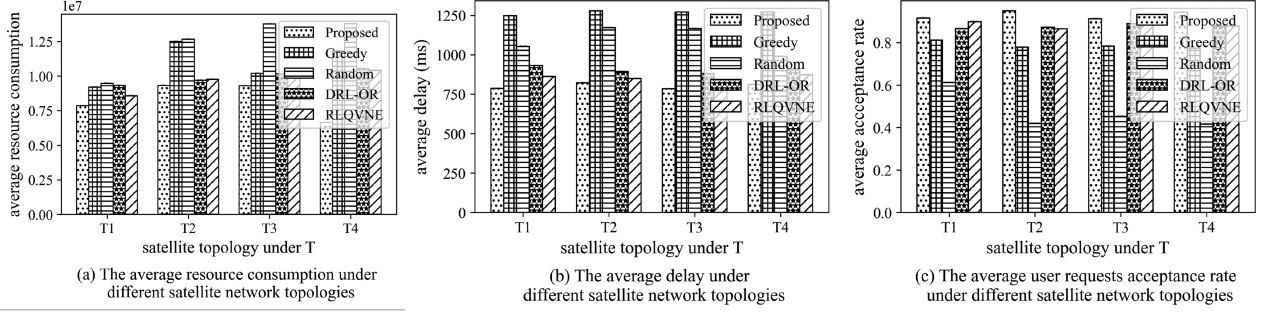


Fig. 8. Performance comparison under different satellite network topologies in terms of average resource consumption, average delay, and average user request acceptance rate.

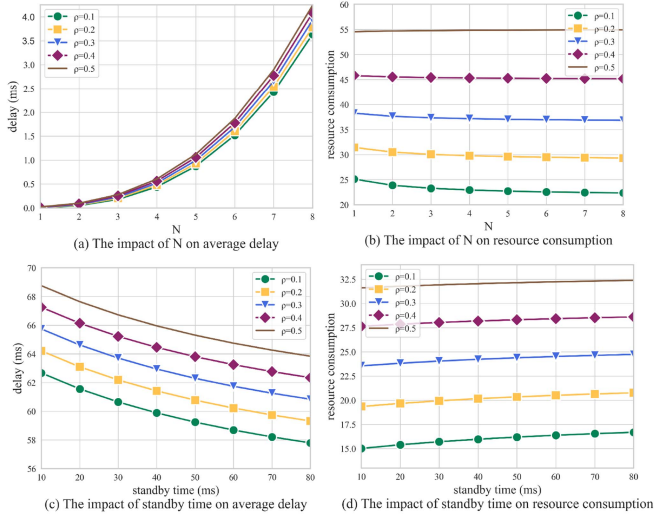


Fig. 9. The impact of switch parameter N and standby duration t_u^{sta} on the average delay and resource consumption of satellite UPF under different user request densities ρ .

phase only occurs after the accumulation of N user requests. Additionally, the standby duration t_u^{sta} also affects the balance between resource consumption and delay.

In this experiment, we set $t_u^{ser} = t_u^{set}$ and configure parameters such as $S(t_u^{set})^2$ and $S(t_u^{ser})^2$. Fig. 9(a) and (b) respectively depict the impact of N on the delay and resource consumption of satellite UPFs. Within a certain range of N , delay exhibits a monotonic increase with the growth of N , and it also increases with higher user request density λ , as expected. It is because a larger N or an increase in the number of user requests implies more users queuing, resulting in increased delay. In terms of resource consumption, as N increases, the resource consumption gradually decreases under different user request densities, which confirms the fact that setting a larger N can save more resources.

The impact of the standby duration t_u^{sta} on delay and resource consumption of satellite UPFs is shown in Fig. 9(c) and (d). As t_u^{sta} increases, satellite UPF delay exhibits a monotonic decrease, while resource consumption shows a monotonic increase. As analyzed earlier, the increase in t_u^{sta} implies a delayed entry into the low-resource consumption close stage, which consequently

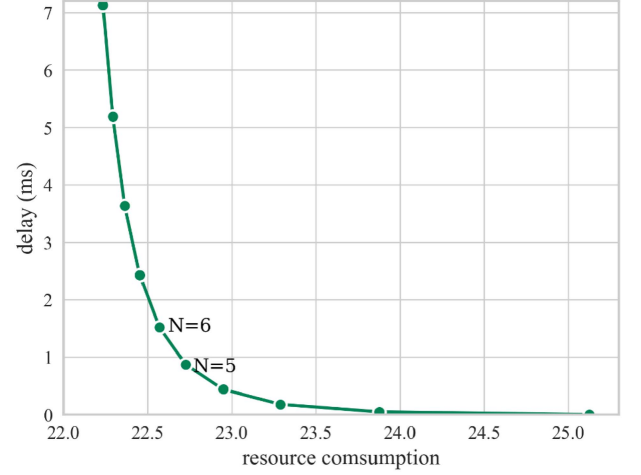


Fig. 10. Delay-resource consumption trade-off with different switch control parameters.

consumes more resources. However, during the standby stage, when a user request arrives, the system immediately enters the service stage without causing additional delays. Therefore, the experimental results presented in Fig. 9(c) and (d) align with real-world expectations.

After investigating the individual parameters N and standby duration t_u^{sta} on the delay and resource consumption of satellite UPFs, we further quantify the delay resource consumption trade-off of satellite UPF switch control, as shown in Fig. 10. It can be observed that excessive pursuit of minimizing delay leads to a sharp increase in resource consumption. Conversely, efforts to reduce resource consumption come at the cost of increased delay. Therefore, when making decisions regarding satellite UPF switch control, it is crucial to carefully consider the trade-off between delay and resource consumption. One viable solution is the flexible adjustment of switch control parameters, which provides valuable insights for the future deployment and operation of satellite UPFs in the industry.

VII. CONCLUSION

The deployment of 5 G core network UPF on satellites marks a firm step towards the integration of 5 G with satellite technology. However, the efficient operation of satellite UPF faces objective

constraints related to service continuity and limited resources. This paper examines the satellite UPF service optimization problem from the perspective of joint optimization of state migration, traffic routing, and satellite UPF switch control. Specifically, the switch control is modeled using an M/G/1 queue model, and the state migration and traffic routing are defined as a target satellite node selection problem. Drawing on reinforcement learning theory, we propose the dynamic switch control parameter adjustment strategy and state-aware satellite UPF service optimization algorithm based on the policy network. Extensive simulation experiments demonstrate that the proposed algorithms notably reduce the delay of satellite UPF switch control and service processes. In the future, we will consider cost-effective strategies for deploying other core network functions on satellites. Additionally, we will continue to explore the impact of different switch control strategies on satellite resource consumption and QoS.

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